

CEM, arbitrariness, and vagueness
Topics in Mind and Language: Conditionals
12 October 2017

1. Recap: arguments we have considered so far for and against CEM

For:

- (i) from certain judgments about objective chance (likelihood)
- (ii) from certain judgments about rational credence
- (iii) from denials and lack of scope distinctions
- (iv) [from quantified conditionals]

Against:

- (i) from the strict conditional analysis
 - which could in turn be argued for on the basis of Reverse Sobel Sequences
- (ii) from 'might'-'would' duality

2. Against CEM: from the similarity account of closeness

Even if we are, like Lewis, pretty liberal about what counts as a comparative similarity relation, it's hard to see how any comparative similarity relation could be antisymmetric.

One could retreat from the view that closeness *is* a comparative similarity relation to the view that there is a comparative similarity relation such that whenever *u* is more similar to *w* than *v* is in that sense, *u* is closer to *w* than *v* is (but not vice versa)!

- Note however that if the Limit Assumption fails for the relevant comparative similarity relation, this sufficient condition means that it's also got to fail for closeness, which means we still can't have CEM. And as Lewis points out, the Limit Assumption fails for many comparative similarity relations.
- Also: the kind of reasoning that drives the chance- and credence-based arguments for CEM fits poorly with the comparative similarity vision.
 - Suppose we tossed a fair coin five times yesterday, starting at noon, and it landed Heads on tosses 3 and 5 and Tails on tosses 1, 2, and 4. Q: How likely was it, at noon, that it would land Heads five times, if it landed Heads at least four times?
 - The kind of reasoning we've been relying on suggests the answer '1/6', based on the fact that there are 6 equiprobable ways for it to land Heads at least four times, of which one involves it landing Heads five times. But if being more similar to actuality is sufficient for being closer to actuality, and the number of coin outcomes matters at all, it looks like the answer's going to be 1/64. So long as the coin doesn't *actually* land Heads every time (something whose chance is 1/64),

some worlds where it lands Heads exactly 4 times will be more similar to actuality than any world where it lands Heads 5 times.

- Tentative conclusion: if you want to hold on to CEM, you had better forget about the idea that closeness has anything to do with similarity. We know more about similarity than we know about closeness—for example, even knowing how the coin landed, we just don't know whether all-Heads worlds are closer than four-Heads worlds (our credence is 1/6).

3. Against CEM: the grounding objection

Consider two coins A and B that have never been tossed. Given CEM, we have the following disjunction:

Either A would have landed Heads if A and B had been tossed and exactly one of them had landed Heads, or B would have landed Heads if A and B had been tossed and exactly one of them had landed Heads

But which one is it? What could *make it the case* that it's this one rather than that one that has the relevant property?

- A time-honored concern! In the late 16th century, Molina and Suarez claimed that God had "middle knowledge" of counterfactuals like 'If this free creature were put in this situation, it would make this choice'. Opponents objected that God could not have such knowledge since there were no facts to be known.

4. Does the grounding objection attack LEM as well as CEM?

So far we have been assuming classical logic, including the validity of the schema LEM:

LEM $A \vee \neg A$

If this is valid, then so its substitution instance

(H) Either Harry is bald, or Harry is not bald

Some philosophers think that (H) must be rejected in the case where Harry is borderline-bald. One of their basic arguments is a kind of grounding objection: what could make it be the case that Harry is bald, or that Harry is not bald? (Or alternatively

To get a better feel for the objection, consider a long row of people all lined up, starting with someone with zero hairs, ending with someone very hirsute, and where each person has just one more hair than his predecessor. Given that person 0 is bald and person 10000000 is not bald, LEM (together with other less controversial rules) leads to the following disjunction: either

Person 0 is bald and person 1 is not bald, or
Person 1 is bald and person 2 is not bald, or ...

...or person 999999 is bald and person 1000000 is not bald.

If this is true, then so is the following claim: there is a last bald person in the sequence. Or alternatively: there is a minimum number of hairs compatible with not being bald. But what could make it the case that any particular number has this status?

But there are also powerful considerations in *favour* of LEM. Most obviously, it can be argued for on the basis of the law of non-contradiction, as follows:

- | | |
|-----------------------------|--------------------------------|
| 1. $\neg(A \wedge \neg A)$ | LNC |
| 2. $\neg A \vee \neg\neg A$ | 1, De Morgan |
| 3. $\neg A \vee A$ | 2, double negation elimination |

So if on reflection we decide we want to keep LEM, there is reason to be suspicious of the grounding objection to CEM.

5. An influential response to the grounding objection to LEM: semantic indecision

Proponents of this view concede something to the grounding objection: although for every property that's entailed by having a billion hairs there is a minimum number of hairs consistent with lacking that property, nothing in our use of the vague word 'bald' singles out any *particular* such property for a unique semantic role. Our use of the word involves **semantic indecision**: there are many different properties all of which count as *admissible interpretations* of the word 'bald'. They have different cutoffs; some of are had by Harry and some are not.

- But many sentences involving the word 'bald' are *supertrue*: they are true on *all* the admissible interpretations. (H) is one example. 'There is a minimum number of hairs consistent with not being bald' is another.
- Supertruth is enough to license assertion once pragmatic factors are cleared away. So we should assert (H), since we should believe *all* the propositions that are its admissible interpretations.
- Some philosophers in this tradition (but not Lewis!) add: truth = supertruth. So a disjunction can be true without having a true disjunct.
 - NB: this is specific to 'true' when applied to *sentences*; it does not carry it over to 'true' as a predicate of *propositions*, or to 'it is true that ...'.
 - We can say the same thing about 'Either it is true that Harry is bald, or it is true that Harry is not bald' that we said about H itself.
- This story applies to all kinds of parts of speech, not just adjectives. In particular it can also apply to *names*, like 'baldness'. So we shouldn't, strictly speaking, be asserting things like 'There is no such thing as baldness', or 'there are multiple properties of baldness', etc. — they are superfalse.

6. Does this suffice as a reply to the grounding objection to CEM?

Supervaluationist response: there are many different binary relations among propositions which are admissible interpretations of 'if' (or for the counterfactual version of 'if?'). For each of these relations R , there is a two place relation A and a three place relation $\leq$, obeying the Limit Assumption and Antisymmetry, such that $R(p, q)$ is true at a world w iff either no P -world bears A to w , or the $\leq_w$ -minimal p -world bearing A to w is a q -world. So instances of CEM are supertrue.

Worry: does this secure the kind of *ignorance* we want?